MATH INQUIRERS HOLIDAY PROGRAM

Question Bank

Three tiers of questions for every lesson, with answers

Notice it. Test it. Explain why.

Question Bank • Ages 10-12 (builds on Primary 5 maths)

Math Inquirers P5 Ages 10-12. Notice it. Test it. Explain why.

How to use this bank

- Each lesson has three tiers: Start here, Think harder and Go further. They are open to everyone. Nobody is placed in a tier: every child can start with Start here, and any child may try Think harder or Go further at any time. Go further questions are good ones to talk about together.
- START HERE questions are approachable for a child who has not yet been taught the topic. They can start with objects, a drawing, counting, guess and check or skills from earlier years, and they use no unexplained words. Use them for warm-ups and to build confidence.
- THINK HARDER questions need the idea taught in the lesson, used in a new setting: two-step problems, patterns with a hidden rule, unknowns and larger numbers.
- GO FURTHER questions need the topic well understood, plus an extension: find every way, work backwards, test a case, justify an answer, change the problem or make up your own. This program is inspired by research-based approaches used in gifted education; it is not an official GEP standard and is not affiliated with MOE.
- Ask for the reason every time: "How do you know?" A correct answer with no reason is only half done.
- Answers are at the back. Do not hand the answer pages to children.

What each lesson practises

Lesson 1	Million Puzzles (Numbers to 10 Million)	6 + 9 + 9 questions
Lesson 2	Fraction Feats (Fractions: Four Operations)	6 + 9 + 9 questions
Lesson 3	Before and After (Fraction Word Problems)	6 + 10 + 9 questions
Lesson 4	Triangle Areas (Area of Triangle)	6 + 8 + 9 questions
Lesson 5	Volume Vault (Volume)	6 + 10 + 9 questions
Lesson 6	Rate Racers (Rate)	6 + 12 + 9 questions
Lesson 7	Percentage Puzzles (Percentage)	6 + 8 + 9 questions
Lesson 8	Angle Chasers (Angles, Triangles and Quadrilaterals)	6 + 10 + 9 questions
Lesson 9	Pattern Generalisers (Patterns and Number Sense)	6 + 8 + 9 questions
Lesson 10	Challenge Day (Mixed Strategies)	6 + 8 + 9 questions

Lesson 1: Million Puzzles

Numbers to 10 Million • Week 1

START HERE (open with objects, drawing or counting)

S1. Write 4 030 205 in words.

S2. What is the value of the digit 6 in 3 621 480?

S3. Round 5 482 716 to the nearest thousand.

S4. Use seven digit cards, 0 1 2 3 4 5 6, each once. Make the greatest 7-digit number you can, and then the smallest. (A 7-digit number cannot start with 0.)

S5. A shop has 48 pens. It sells 6 packs. Each pack has 3 red pens and 4 blue pens. How many pens are left? (Draw the packs or use counters if it helps.)

S6. Estimate 6912 × 48 by rounding 6912 to the nearest thousand and 48 to the nearest ten.

THINK HARDER (the lesson idea in a new setting)

T1. How many 7-digit numbers have digits that add up to 2? List them all.

T2. Put ONE pair of brackets into 2 + 3 × 4 + 5 to make 25. Then put one pair in a different place to make 29.

T3. A 7-digit number rounded to the nearest thousand is 2 500 000. Its digits add up to 9. Write the range first (the smallest and the greatest number that round to 2 500 000). Then find ALL the numbers that fit. How do you know you have them all?

T4. Use the seven digit cards 1, 2, 5, 6, 7, 8, 9 once each to make the 7-digit number closest to 7 000 000. Test both sides of the target.

T5. A whole number rounded to the nearest thousand is 640 000. What is the greatest it could be? The smallest?

T6. Use each of the digits 1, 2, 3, 4, 5, 6 and 7 once. What is the difference between the greatest and the smallest 7-digit number you can make?

T7. Which is the best estimate of 4108 × 29: 12 000, 120 000 or 1 200 000? Explain.

T8. Use 2, 3, 5 and 7 once each, with +, −, ×, ÷ and brackets, to make 29, then 31, then 100.

T9. Use four 4s with +, −, ×, ÷ and brackets to make 3, then 5, then 7.

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Make up your own digit riddle for a 7-digit number. Use at least 5 clues (for example: all digits different, a rounding clue, a clue that compares two digits, a digit sum). Only ONE number may fit. Swap with a partner: do you get the same answer? How can you be sure your riddle has only one answer?

G2. Write 1 000 000 as the product of two whole numbers that have no zero digits at all.

G3. Ben multiplies a 4-digit number by a 3-digit number. How many digits can his answer have?

G4. A palindrome reads the same forwards and backwards, like 3 456 543. How many 7-digit palindromes are there between 3 000 000 and 4 000 000?

G5. Use 1, 2, 3, 4, 5 in this order, with +, −, ×, ÷ and brackets, to make 100.

G6. A heart beats about 70 times a minute. About how many days does it take to beat 1 000 000 times?

G7. Put brackets into 1 + 2 × 3 + 4 × 5 + 6 to make the answer as big as possible. You may use more than one pair. What is the biggest answer?

G8. I am a 7-digit odd number. My digits are all different and add up to 42. Rounded to the nearest hundred thousand, I am 7 000 000. I am the smallest number that fits. Who am I?

G9. Ali writes all the whole numbers from 1 to 1000. How many digits does he write? (Go further: how many if he writes all the numbers from 1 to 1 000 000?)

Lesson 2: Fraction Feats

Fractions: Four Operations • Week 1

START HERE (open with objects, drawing or counting)

S1. 4 friends share 3 chocolate bars equally. Draw the bars and cut them. How much chocolate does each friend get?

S2. Siti ate 1/2 of a pizza and Ben ate 1/4 of the same pizza. Draw the pizza. What fraction did they eat altogether? What fraction is left?

S3. A pizza is cut into 12 equal slices (draw it). Ravi eats 1/3 of the pizza and Mei eats 1/4 of the pizza. How many slices did they eat? What fraction of the pizza is left?

S4. Find 3/5 of 15 sweets. (Draw 15 dots and put them into 5 equal groups.)

S5. Half of a cake is left. Ali eats half of what is left. Draw the cake. What fraction of the WHOLE cake did Ali eat?

S6. Mei has 3/4 of a pizza and Ali has 2/3 of a pizza (the pizzas are the same size). Who has more? Draw two bars to show how you know.

THINK HARDER (the lesson idea in a new setting)

T1. Write 3/8 as the sum of two different unit fractions in TWO different ways. How do you know when to stop looking?

T2. Ben ate 1/3 of a pizza. Ali ate 1/4 of what was left. What fraction of the pizza is left?

T3. 1/2 − 1/3 = 1/6 and 1/3 − 1/4 = 1/12. Use the pattern to work out 1/4 − 1/5 and 1/9 − 1/10.

T4. 4 litres of juice are poured equally into 6 bottles. How much juice is in each bottle?

T5. Which is bigger, 5/6 or 6/7? Explain without finding common denominators.

T6. (A puzzle that goes further) A rope 3/4 m long is cut into 3 equal pieces. How long is each piece? Draw a bar to help.

T7. Find a fraction between 2/5 and 1/2 with 20 as the bottom number.

T8. Which is bigger, 7/8 of 24 or 5/6 of 30?

T9. Use the digit cards 1, 2, 3 and 4 once each in the boxes to make two fractions and add them (fractions need not be proper). Which sum is closest to 1? Test every way.

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Write 1 as the sum of three different unit fractions. Find all the ways.

G2. Work out 1/2 × 2/3 × 3/4 × 4/5 × … × 99/100.

G3. Mrs Tan baked some cookies. She gave 1/2 of them to her neighbours, 1/3 of the remainder to Ben and 1/4 of what was then left to Mei. She had 30 cookies left. How many did she bake?

G4. Make up your own story about a fraction of a REMAINDER (it must use "of what was left" at least once) with a whole-number answer. Solve it with a bar model, swap with a partner, and check each other. Then change one fraction in your story: what changes in the answer, and what stays the same?

G5. Use the digit cards 1, 3, 5 and 7 once each to make two fractions. Take the smaller from the bigger. What is the smallest difference you can make (more than 0)?

G6. Write 3/4 as the sum of three DIFFERENT unit fractions. Find all the ways and explain how you know you have found them all.

G7. Work out 1/6 + 1/12 + 1/20 + 1/30 + 1/42 without finding a common denominator for all five. (Hint: 1/6 = 1/2 − 1/3.)

G8. 5 pizzas are shared equally among 12 children. Each child must get exactly 2 pieces, and each piece must be a unit fraction of a pizza (like 1/3 or 1/4). How can the pizzas be cut?

G9. A tank is full of water. Every hour, 1/3 of the water that is in the tank is let out. After 3 hours there are 16 litres left. (a) How much water was in the full tank? (b) After how many hours is less than half of the original water left? (c) How much water has been let out by then?

Lesson 3: Before and After

Fraction Word Problems • Week 2

START HERE (open with objects, drawing or counting)

S1. Ali has 18 counters and Mei has 6 counters. Ali gives Mei 4 counters. How many counters does each child have now? What is the total before, and what is the total after? (Use counters or draw dots.)

S2. Ben is 10 years old and his father is 38. How old will each of them be in 5 years? How many years older than Ben is his father now? And in 5 years?

S3. Siti had $60. She spent 1/4 of it on a book and 1/3 of the remainder on a T-shirt. How much money did she have left?

S4. Ali had 26 stamps and Ravi had 34. How many stamps must Ravi give Ali so that they have the same number?

S5. There are 40 pupils in a hall. 3/8 of them are boys. How many more girls than boys are there?

S6. A jar has 24 red beads and 12 blue beads. What fraction of the beads are blue? Then 12 more red beads are put in. What fraction of the beads are blue now? Which colour did not change?

THINK HARDER (the lesson idea in a new setting)

T1. Ravi spent 1/4 of his money on Monday and 2/5 of the remainder on Tuesday. He spent $66 altogether. How much money did he have at first?

T2. A box had red beads and blue beads. 3/7 of the beads were red. Then 30 red beads were added. Now 3/5 of the beads are red. How many beads were in the box at first? How many are in the box now?

T3. Mrs Tan is 41 years old and her son is 9. How many years ago was her son's age 1/5 of her age?

T4. Ben and Siti had $140 altogether. Ben spent 1/2 of his money and Siti spent $20. Then they had the same amount left. How much did Ben have at first?

T5. Ali had 40 more stickers than Mei. Each of them was given 10 more stickers. Then Mei had 3/5 as many stickers as Ali. How many stickers did Mei have at first?

T6. A tank was 3/4 full of water. After 18 litres were used, it was 1/2 full. How many litres does the tank hold when it is full?

T7. Siti gave 1/3 of her stickers to Mei. Then they had the same number of stickers. Mei had 15 stickers at first. How many did Siti have at first?

T8. Ravi had $45 and Ben had $15. Their aunt gave each of them the same amount of money. Then Ben had 1/2 as much as Ravi. How much did the aunt give each boy?

T9. A box had red pens and blue pens. 2/3 of the pens were blue. After 12 blue pens were taken out, 1/2 of the pens were blue. How many pens were in the box at first?

T10. Ali has 50 marbles and Ben has 30 marbles. How many marbles must Ali give Ben so that Ali has 3/5 as many marbles as Ben?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Make up your own before-and-after story in which the total stays the same and a fraction is used in the After (like "Ali has 1/2 as many as Ben"). Solve it with a model, swap with a partner and check each other. What if the fraction in your story were changed?

G2. Ali has 18 counters and Mei has 6. Ali may give Mei some counters (he can give none, or up to all 18). After the giving, one child has exactly 2, 3, 4 … times as many counters as the other (the two numbers are not equal). List ALL the ways. How do you know you have them all?

G3. A father and his son are 50 years old altogether. In 5 years, the son's age will be 1/3 of the father's age. How old is the son now?

G4. Ali had 3/5 as many cards as Ben. After Ben gave Ali 20 cards, they had the same number. How many cards did they have altogether?

G5. In a basket, 1/3 of the fruits were apples. 10 apples were eaten and 10 pears were put in. Now 1/6 of the fruits are apples. How many fruits are in the basket?

G6. Siti spent 2/5 of her money on a dress and 1/3 of the remainder on shoes. The dress cost $30 more than the shoes. How much money did Siti have at first?

G7. Mei had $200 and Siti had $80. Mei spent 3 times as much money as Siti. Then they had the same amount left. How much money did Siti spend?

G8. Ravi had some stickers. He gave 1/2 of them and 4 more to Ali. Then he gave 1/2 of the stickers he had left and 4 more to Ben. He had 4 stickers left. How many stickers did he have at first?

G9. Ali and Ben had 140 stamps altogether. Ali gave away 1/3 of his stamps and Ben gave away 1/2 of his. Then they had the same number of stamps left. How many stamps did Ben give away?

Lesson 4: Triangle Areas

Area of Triangle • Week 2

START HERE (open with objects, drawing or counting)

S1. On 1-cm grid paper draw a rectangle 6 cm by 4 cm. Cut it along a diagonal into two triangles. What is the area of each triangle? (Fold or cut to check that the two triangles are the same.)

S2. A square has sides of 8 cm. Draw it. Put a point anywhere on the top side and join it to the two bottom corners. Cut the three pieces out: do the two side pieces fit exactly on the middle triangle? What is the area of the middle triangle?

S3. On grid paper draw a right-angled triangle with a base of 4 cm and a height of 3 cm. Draw the same triangle again, turned upside down, so that the two triangles make a rectangle. What is the area of ONE triangle?

S4. Draw a rectangle 5 cm by 4 cm. Mark a point on the top side. Join it to the two bottom corners. What is the area of the middle triangle? Try a different point: does the area change?

S5. Two paper triangles are each exactly half of a 6 cm by 5 cm rectangle. What is the total area of the two triangles?

S6. On grid paper draw a rectangle 8 squares long and 3 squares tall. Shade a triangle that has the long side as its base and its top corner anywhere on the opposite side. Compare with your partner: do your shaded areas match? What is the area?

THINK HARDER (the lesson idea in a new setting)

T1. A triangle has an area of 45 cm². Its base is 10 cm. What is its height?

T2. A right-angled triangle has sides of 12 cm, 16 cm and 20 cm. What is its area?

T3. A triangle and a square have the same area. The square has sides of 6 cm. The triangle has a base of 9 cm. What is the height of the triangle?

T4. A triangle and a rectangle have the same base, 12 cm, and the same height, 7 cm. What is the difference between their areas?

T5. Triangle P has a base of 8 cm and a height of 5 cm. Triangle Q has a base of 4 cm and the same area as P. What is the height of Q?

T6. A rectangle is 12 cm by 8 cm. M is the middle of the top side. M is joined to the two bottom corners. What is the area of the triangle? What is the area of the rest of the rectangle?

T7. A figure is a square of side 6 cm with a triangle on top. The base of the triangle is the top side of the square and its height is 4 cm. What is the area of the figure?

T8. Ali draws a triangle with a base of 10 cm and a height of 6 cm. Mei draws a triangle with a base of 5 cm and twice the area of Ali's triangle. What is the height of Mei's triangle?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Make up your own shaded-area puzzle. Draw a rectangle with the lengths of its sides, mark points on the sides and join them to make a shaded triangle and two or three white triangles. Work out the shaded area in TWO ways (split it, or take the white triangles away) and check that they agree. Swap puzzles with a partner.

G2. A rectangle is 12 cm by 8 cm. P is the middle of the top side and Q is the middle of the right side. What is the area of the triangle joining P, Q and the bottom-left corner?

G3. A triangle has a base of 12 cm and a height of 5 cm. (a) Its base is made 3 times as long and the height stays the same. What is the new area? (b) The base is made 3 times as long AND the height is made half as long. What is the new area? (c) The base is made 4 times as long. How must the height change to keep the area the same? (d) Say in words what happens to the area when the base and the height are both multiplied.

G4. Square ABCD has sides of 8 cm (A top-left, B bottom-left, C bottom-right, D top-right). BC is extended to E so that CE = 4 cm. AE cuts CD at F. What is the difference between the area of triangle ADF and the area of triangle FCE?

G5. List every pair of whole numbers (base and height, in cm) for a triangle with an area of 6 cm².

G6. A triangle has sides of 13 cm, 14 cm and 15 cm. The height that meets the 14 cm side at a right angle is 12 cm. (a) What is the area of the triangle? (b) What is the height that meets the 15 cm side at a right angle? (c) Why is it shorter than 12 cm?

G7. A right-angled triangle has sides of 15 cm, 20 cm and 25 cm. What is its height when the 25 cm side is the base?

G8. Two squares stand side by side on a line, touching: a big square of side 10 cm and a small square of side 6 cm. Join the bottom-left corner of the big square, the top-right corner of the big square and the top-right corner of the small square to make a triangle. What is its area? Try again with a small square of side 4 cm. What do you notice?

G9. Triangle ABC has an area of 54 cm². D is a point on BC with BD = 4 cm and DC = 8 cm. What is the area of triangle ABD?

Lesson 5: Volume Vault

Volume • Week 3

START HERE (open with objects, drawing or counting)

S1. Build a box 3 cubes long, 2 cubes wide and 2 cubes high with 1-cm cubes (or draw it layer by layer). How many cubes are in the bottom layer? How many layers are there? How many cubes are used altogether?

S2. Use 12 cubes to build a flat rectangle, one cube high. How many different rectangles can you make? (Turning a rectangle round does not make a new one.) Write the length and width of each.

S3. A tank has a base of 10 cm by 10 cm. If the water is 1 cm deep, there are 100 one-cm cubes of water. How many cubes of water are there if the water is 3 cm deep? 10 cm deep? How many litres is the water when it is 10 cm deep? (1 ℓ = 1000 cm³.)

S4. How many 1-cm cubes (dice) fill a box 5 cm long, 4 cm wide and 3 cm high? Use dice, or draw one layer and count the layers.

S5. A tank has a base 4 cm by 3 cm. The water is 2 cm deep. Twelve 1-cm cubes are dropped in and sink to the bottom. How much higher does the water rise? (Think of the cubes as one more layer of water.)

S6. A bottle holds 2 litres of water. How many cm³ is that? How many ml is 2.5 ℓ? (1 ℓ = 1000 ml = 1000 cm³.)

THINK HARDER (the lesson idea in a new setting)

T1. How many cubes with 2-cm edges can be packed in neat layers into a box 9 cm long, 6 cm wide and 5 cm high?

T2. A tank is 1/3 full of water. After 12 ℓ more is poured in, it is 3/5 full. How many litres does the tank hold when full?

T3. A cuboid has a square base and a height of 5 cm. Its volume is 245 cm³. How long is each side of the base?

T4. A cuboid 3 cm by 4 cm by 5 cm is built from 1-cm cubes and painted on the outside. How many cubes have no paint at all?

T5. A tank is 60 cm long, 40 cm wide and 50 cm high. A bucket holds 8 ℓ. What is the fewest number of buckets of water needed to fill the tank at least 3/4 full?

T6. A tank has a base 40 cm by 30 cm. It holds 36 ℓ of water. How deep is the water?

T7. A cube-shaped container with 20-cm edges is full of water. All of it is poured into an empty tank with a base 50 cm by 32 cm. How deep is the water in the tank?

T8. A tank with a base 50 cm by 20 cm has water 8 cm deep. Ali pours in 3 ℓ of water. Then he drops in a stone with a volume of 500 cm³, and it sinks. How deep is the water now?

T9. A tank has a base 40 cm by 25 cm and the water is 20 cm deep. Ali drops in a stone and the water rises another 3 cm. What is the volume of the stone?

T10. Container A has a base 30 cm by 20 cm and is 24 cm high; it is full of water. Tank B has a base 40 cm by 30 cm and is empty. All the water from A is poured into B. Then some water is poured back from B into A until the water in A and B is equally deep. How deep is it?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. A big cube is painted on the outside and cut into small cubes of the same size. 96 small cubes have exactly one painted face. How many small cubes are there altogether?

G2. How many different cuboids can be built with exactly 64 one-centimetre cubes, using all of them each time?

G3. A tank has a base 40 cm by 30 cm. A cube with edges of 10 cm sits on the bottom, and the water is 6 cm deep around it. How much water is in the tank?

G4. Water in a tank is 15 cm deep. When 4 identical bricks are put in, the water rises to 18 cm. 6 more of the same bricks are put in. How deep is the water now? (The bricks stay under water.)

G5. Make up your own tank puzzle with a stone. Choose the size of the base, the volume of the stone and how the water changes, then give your partner the puzzle. Check your partner's answer. What if the base were twice as wide: how would the rise change?

G6. Bricks measure 2 cm by 3 cm by 4 cm. They are all placed the same way up to build a solid cube. What is the fewest number of bricks needed?

G7. Three faces of a cuboid have areas 12 cm^2, 15 cm^2 and 20 cm^2. What is the volume of the cuboid?

G8. A cuboid 3 cm by 4 cm by 5 cm is built from 1-cm cubes and painted on the outside. How many cubes have exactly 2 painted faces? How many have exactly 1?

G9. A tank has a base 30 cm by 20 cm and water 10 cm deep. A block 10 cm by 10 cm by 15 cm is stood upright on the bottom. How deep is the water now?

Lesson 6: Rate Racers

Rate • Week 3

START HERE (open with objects, drawing or counting)

S1. 4 apples cost $2. How much do 8 apples cost? How much do 12 apples cost? How much does 1 apple cost? (Make a table if it helps.)

S2. A printer prints 24 pages a minute. How long does it take to print 360 pages?

S3. Pack A has 3 pens for $2. Pack B has 6 pens for $5. Which is the better buy? (Hint: how much would 6 pens cost if you bought Pack A twice?)

S4. A tap fills a 120 ℓ tank at 8 ℓ a minute. How long does it take?

S5. Tap A fills 2 litres every minute and tap B fills 3 litres every minute. Both are turned on together. How many litres are in the tank after 1 minute? After 2 minutes? After 5 minutes? How long do they take to put in 20 litres?

S6. Nuts cost $1.50 for every 100 g. How much do 650 g cost?

THINK HARDER (the lesson idea in a new setting)

T1. A tap fills a tank in 12 minutes at 15 ℓ a minute. When a second tap is also turned on, the tank fills in 9 minutes. How many litres a minute does the second tap fill?

T2. Eggs are sold in boxes of 10 for $3.30 and boxes of 6 for $2.10. Siti needs at least 28 eggs. What is the least she can pay?

T3. A car park charges $1.50 for the first hour and $1 for every half hour or part of a half hour after that. How much does it cost to park from 10.15 a.m. to 2.50 p.m.?

T4. Phone plan A costs $15 a month plus 10¢ a minute. Plan B costs 25¢ a minute with no monthly fee. For how many minutes in a month do the two plans cost the same?

T5. 3 machines make 360 masks in 4 minutes. At the same rate, how many masks do 5 machines make in 6 minutes?

T6. Shop X sells pens at 80¢ each. Shop Y sells 4 pens for $3.00 and gives 1 extra pen free with every 4 pens bought. Mei needs 20 pens. Which shop is cheaper, and by how much?

T7. A full tank holds 300 ℓ. A drain lets out 15 ℓ a minute, but a tap is left on that adds 5 ℓ a minute. How long does the tank take to empty?

T8. A dripping tap drips 5 ml of water every 2 seconds. How many litres of water does it waste in one whole day?

T9. Three shops sell the same can of drink: Shop A, 6 cans for $4.20; Shop B, 4 cans for $2.92; Shop C, a deal of 12 cans for $7.80 (10 cans and 2 free). Ben needs exactly 20 cans. He may buy from more than one shop, but only whole packs. Find the cheapest way and its cost.

T10. Printer P prints 30 pages a minute and printer Q prints 20 pages a minute. Mei splits a 900-page job between them so that both printers start together and finish at the same time. How many pages does she give to P?

T11. The 240 ℓ tank is full. Both taps are off, but the crack still lets out 6 ℓ a minute. How long until the tank is half full?

T12. A car park charges $1.20 for the first hour and 60¢ for every half hour (or part of a half hour) after that. Mei pays $3.60. What is the longest time she could have parked?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Printer P prints a job in 6 minutes. Printer Q prints the same job in 3 minutes. How long do they take working together?

G2. Siti buys pens at 3 for $2 and sells them at 2 for $2. How many pens must she buy and sell to make a profit of $10?

G3. A machine starts at 9.52 a.m. and makes 50 cups a minute. At 10.00 a.m. a second machine starts, making 30 cups a minute. At what time have they made 1600 cups altogether?

G4. Ben needs at least 20 cans. Shop A sells 6 cans for $4.20, shop B sells 4 cans for $2.92 and shop C sells a deal of 10 cans for $7.80 with 2 cans free. He can buy from more than one shop, but only whole packs. What is the least he can pay?

G5. A tap fills 9 ℓ a minute. How long does it take to fill an empty tank 60 cm long, 50 cm wide and 45 cm high?

G6. Make up your own better-buy puzzle: three pack sizes of the same snack where the BIGGEST pack is NOT the best buy. Give the sizes and prices, swap with a partner, and ask them to find the best buy and prove it. What price would make the biggest pack the best buy?

G7. 3 cats eat 3 fish in 3 minutes. At this rate, how many cats are needed to eat 100 fish in 100 minutes?

G8. Candle A is 20 cm tall and burns down 2 cm every hour. Candle B is 14 cm tall and burns down 1 cm every hour. They are lit at the same time. After how many hours are they the same height? How tall are they then?

G9. A tap fills an empty bath in 12 minutes. With the plug out, a full bath empties in 20 minutes. Ben turns on the tap but forgets the plug. How long does the bath take to fill?

Lesson 7: Percentage Puzzles

Percentage • Week 4

START HERE (open with objects, drawing or counting)

S1. A hundred-square has 100 small squares. 35 of them are shaded. What percentage is shaded? What percentage is NOT shaded? (Percent means “out of 100”.)

S2. Percent means “out of 100”, so 50% is a half. Find 50% of $80. Then use it to find 25% of $80 (half of 50%) and 10% of $80 (one tenth).

S3. A book costs $50. It has 10% off, which is one tenth of the price. How much money is taken off? What do you pay?

S4. 24 of the 40 pupils in a class walk to school. What percentage walk to school?

S5. Ali had $60. He spent 25% of it. How much did he have left?

S6. A notebook costs $20 before GST. GST is 9%. 1% of $20 is 20¢. How much is the GST? How much do you pay altogether?

THINK HARDER (the lesson idea in a new setting)

T1. At a hawker centre, a plate of chicken rice went up from $4 to $5. A month later, it went back down from $5 to $4. (a) Find the percentage increase from $4 to $5. (b) Find the percentage decrease from $5 to $4. (c) Both changes are $1. Why are the two percentages different? Hint: percentage change = the change out of the amount you START with.

T2. A box of cereal costs $15 in two shops. Shop X: "Buy 3, get 1 free!" Shop Y: "30% off every box!" Mei buys 4 boxes. (a) How much does she pay at each shop? Which is the better buy, and by how much? (b) Shop X's deal is really how many percent off?

T3. Siti says, "15% of $80 is $8 + $4 = $12." Explain her method. Use it to find 15% of $46.

T4. A $250 bicycle is sold at a 12% discount. How much money is saved? What is the sale price?

T5. A watch costs $80 before GST. GST is 9%. Mei pays with a $100 note. How much change does she get?

T6. Siti had $200. She spent 40% of it on shoes and 50% of the rest on a bag. How much money did she have left?

T7. Which is bigger: 30% of 70 or 70% of 30? Try another pair like this. What do you notice?

T8. Ravi scored 18 out of 25 in Test A and 27 out of 40 in Test B. In which test did he do better?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Make up your own percentage trap: a story in which a price goes up by a percentage and then down by a percentage (or two percentages are added) and someone gets the wrong answer. Use real numbers to show what really happens. Swap with a partner: can they find the mistake? What if the two percentages were different?

G2. After a 20% discount, a shirt costs $40. Ben says the price before the discount was $48. Is he right? Find the price before the discount.

G3. A tank is 40% full. After 60 ℓ of water is poured in, it is 70% full. How much water can the tank hold?

G4. Mei's score went up by 20% from Test 1 to Test 2. Then it went down by 20% from Test 2 to Test 3. Her Test 3 score was 48. What was her Test 1 score?

G5. A school has 800 pupils. 45% are boys. 20% of the boys and 30% of the girls take the school bus. What percentage of all the pupils take the bus?

G6. A box has 20 marbles. 75% of them are red and the rest are blue. How many red marbles must be taken out so that only 50% of the marbles left are red?

G7. A price tag says $109, GST included. GST is 9%. Ali says the GST is 9% of $109, which is $9.81. Is he right? How much GST is in the price?

G8. An $80 lamp is on sale in two shops. Shop P takes 25% off. Shop Q takes 10% off, and then a further 15% off the new price. In which shop is the lamp cheaper, and by how much?

G9. Mrs Tan spent 40% of her money on groceries and 25% of the remaining money on a gift. She had $54 left. How much money did she have at first?

Lesson 8: Angle Chasers

Angles, Triangles and Quadrilaterals • Week 4

START HERE (open with objects, drawing or counting)

S1. Draw any triangle on paper and tear off its three corners. Put the three corners side by side with their points touching. What do you notice? What do the three angles add up to? (A straight line is a half-turn, 180°.)

S2. Two angles sit side by side on a straight line. A straight line is 180°. One of the angles is 127°. What is the other angle?

S3. A round pizza is cut into 8 equal slices. A full turn is 360°. What is the angle of one slice at the centre? What is the angle of 2 slices together?

S4. Two angles of a triangle are 48° and 67°. Draw the triangle with a ruler and a protractor and measure the third angle. What do all three angles add up to?

S5. The corner of a sheet of paper is 90°. Fold the corner exactly in half. What angle do you make? Fold it in half again: what angle now?

S6. A rectangle has four corners of 90°. Draw any four-sided shape (not a rectangle), tear off its four corners and fit them together round a point. What do the four angles add up to?

THINK HARDER (the lesson idea in a new setting)

T1. In parallelogram ABCD, angle A is 3 times angle B. Find angle A and angle B. Give your reason.

T2. An isosceles triangle has an angle of 110°. Find the other two angles. Why is there only one answer?

T3. ABCD is a parallelogram and E is a point on DC with DA = DE. Angle ABC = 56°. (The figure is the same one as in Start here Q4.) Find angle AEC and angle EAB. Write the reason for each step.

T4. ABCD is a square. CDE is an equilateral triangle, and E is outside the square. Find angle ADE and angle DAE.

T5. In trapezium ABCD, AB is parallel to DC. Angle DAB is 84° and angle ABC is 97°. Find angle ADC and angle BCD.

T6. Rhombus ABCD has angle ABC = 100°. The diagonal AC is drawn. Find angle BAC.

T7. What is the smaller angle between the hands of a clock at 5 o'clock? At 8 o'clock?

T8. How many degrees does the minute hand of a clock turn through in 10 minutes? In 45 minutes?

T9. In a parallelogram ABCD, angle A is 40° bigger than angle B. Find angle A and angle B.

T10. Two straight lines cross. One of the four angles is 38°. What are the other three? How many different sizes are there?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Two identical right-angled triangles with angles 30°, 60° and 90° are joined with their shorter sides (the sides opposite the 30° angles) touching, with the right angles together, to make one big triangle. What are the angles of the big triangle?

G2. Make up your own angle-chase puzzle: draw a figure with a triangle and a parallelogram (or a rhombus) sharing a side, give just two or three angles, and ask for an angle that needs at least THREE steps. Write the reason for every step in your answer. Swap with a partner and check each other. What if you changed one of the given angles?

G3. In triangle ABC, angle B = 70° and angle C = 50°. The line AD cuts angle A into two equal angles, with D on BC. Find angle ADB and angle ADC.

G4. An isosceles triangle has angles that are all whole numbers of degrees, and one of them is 20°. List every possible triangle.

G5. Triangle ABC has angle A = 20°, angle B = 60° and angle C = 100°. D is a point on AB so that angle ACD = 40°. Find the three angles of triangle ACD and the three angles of triangle BCD.

G6. In parallelogram ABCD, angle ABC = 72°. E is a point on AD with AB = AE. Find angle ABE and angle EBC.

G7. ABC is an isosceles triangle with AB = AC and angle BAC = 40°. D is a point on AC with BD = BC. Find angle ABD.

G8. ABCD is a square. E is a point inside it so that ABE is an equilateral triangle. Find angle DEC.

G9. In an isosceles triangle, one angle is 30° bigger than another angle. Find all the possible sets of three angles.

Lesson 9: Pattern Generalisers

Patterns and Number Sense • Week 5

START HERE (open with objects, drawing or counting)

S1. 5, 9, 13, 17, … What is the 10th number in this pattern?

S2. Matchstick squares in a row use 4, 7, 10, … sticks. How many sticks are needed for 8 squares?

S3. 2, 6, 10, 14, … Is 50 in this pattern? Explain.

S4. There are 15 bicycles and tricycles in a shed. They have 37 wheels altogether. How many tricycles are there?

S5. The rule for a dot pattern is: number of dots = 3 × figure number + 2. How many dots are in figure 15?

S6. Pens cost $2 each and files cost $3 each. Ben buys 10 items and pays $24. How many files does he buy?

THINK HARDER (the lesson idea in a new setting)

T1. Beads repeat in the order red, red, blue, yellow, yellow, yellow. What colour is the 50th bead? How many yellow beads are in the first 100?

T2. 40 square tables in one long row seat 2 × 40 + 2 = 82 people. How many can sit if the tables are put in 8 rows of 5 tables instead?

T3. A farm has chickens and goats. There are 35 heads and 94 legs. How many goats are there?

T4. Work out 1 + 3, 1 + 3 + 5 and 1 + 3 + 5 + 7. What do you notice? Use it to add the first 20 odd numbers, 1 + 3 + … + 39.

T5. Ali saves $3 in week 1. Each week he saves $2 more than the week before. How much does he save in week 15?

T6. 3, 8, 13, 18, … What is the first number in this pattern that is greater than 200?

T7. Two whole numbers next to each other multiply to give 702. What are they?

T8. A test has 10 questions. A correct answer scores 3 marks and 1 mark is taken away for each wrong answer. Siti answered every question and scored 18. How many did she get correct?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. A staircase is built from squares: step 1 is 1 square, step 2 adds a column of 2 (3 squares altogether), step 3 adds a column of 3 (6 squares). How many squares are there in the staircase with 10 steps?

G2. Make up your own growing pattern with matchsticks, dots or tiles. Draw the first three figures, work out the rule in words, and write two questions for a partner: the number of sticks in figure 50, and which figure uses exactly 100 sticks (or explain why no figure does). Swap and check. What if the pattern started with a different first figure?

G3. How many squares of all sizes are there on a 4 by 4 grid?

G4. Spiders have 8 legs and beetles have 6 legs. A box holds 20 of them. (a) If there are 8 spiders, how many legs are there altogether? (b) List ALL the possible totals of legs. (c) Mei says the box has 135 legs. Why must she be wrong? How many spiders would give 148 legs?

G5. How many digits are needed to number the pages of a book from page 1 to page 150?

G6. A square grid of matchsticks: a 1 by 1 grid uses 4 sticks and a 2 by 2 grid uses 12 sticks. How many sticks does a 10 by 10 grid use?

G7. A dot pattern makes diamonds. Figure 1 has 1 dot. Figure 2 has rows of 1, 3 and 1 dots (5 dots). Figure 3 has rows of 1, 3, 5, 3 and 1 dots (13 dots). How many dots are in figure 6?

G8. A quiz has 25 questions. A correct answer scores 4 marks, 1 mark is taken away for each wrong answer, and a blank scores 0. Mei left 3 questions blank and scored 68. How many did she get correct?

G9. Ali writes the counting numbers one after another without spaces: 123456789101112131415... What is the 100th digit he writes?

Lesson 10: Challenge Day

Mixed Strategies • Week 5

START HERE (open with objects, drawing or counting)

S1. A number is multiplied by 4, then 6 is taken away, and the answer is 50. What was the number?

S2. Ali has 1/2 of his money left after buying a book for $9. How much did he have at first?

S3. Two numbers add up to 40 and the bigger is 8 more than the smaller. What are the two numbers?

S4. Ravi, Siti and Ben each have a different pet: a rabbit, a hamster and a bird. Ravi has no bird and no hamster. Siti has no hamster. Who has the bird?

S5. How many squares of all sizes are in a 2 by 2 grid of small squares?

S6. Pens cost $2 and pads cost $3. Ben buys 10 items and pays $24. How many pads?

THINK HARDER (the lesson idea in a new setting)

T1. Siti spent 2/5 of her money, then 1/3 of what was left. She had $24 left at the end. How much did she start with?

T2. How many 3-digit numbers have digits that add up to 5? List them all in order.

T3. The sum of two consecutive numbers (numbers that are next to each other, like 40 and 41) is 145. What are the two numbers?

T4. A snail climbs 3 m up a 10 m wall each day, and slips down 2 m each night. On which day does it reach the top?

T5. Ben has 14 coins, all 50¢ and 20¢ coins, worth $4.90. How many of each?

T6. What is the smallest whole number bigger than 1 that leaves a remainder of 1 when divided by 2, by 3, by 4, by 5 and by 6?

T7. I think of a number, multiply it by 4, take away 10, halve it and add 3. The answer is 22. What is my number?

T8. 1/2 of a number is 5 more than 1/3 of the same number. What is the number?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. At the first bus stop, 1/3 of the passengers got off. At the second stop, half of those left got off and 3 got on. There are now 12 passengers. How many were on the bus at first?

G2. How many rectangles of all sizes are there in a 1 by 5 strip of 5 small squares?

G3. In the 3 by 4 grid of Question 4 there are 20 squares. How many rectangles of ALL sizes are there (a square counts as a rectangle)? Count by size. Then explain how you know you have not missed any or counted any twice.

G4. How many times does the digit 4 appear when you write all the whole numbers from 1 to 50?

G5. Make up your own logic puzzle with four people and four things (pets, dishes, sports ...). Write at least four clues. Test it on a partner: does it have exactly ONE answer? Which clue could you take away without spoiling it, and which clue could you NOT take away?

G6. A monkey eats half of the rambutans in a pile and 1 more, every day for 3 days. After the third day there are none left. How many were in the pile at the start?

G7. A water tank is 1/3 full. After 20 litres are added it is 1/2 full. Then 1/4 of the water in the tank is let out. How many litres of water are in the tank now? What fraction of the tank is full?

G8. 25 coins of 50¢, 10¢ and 20¢ are worth $5.30 altogether. There are twice as many 10¢ coins as 50¢ coins. How many coins of each kind are there?

G9. Ali, Ben, Mei and Siti sit in a row of 4 chairs. Mei is not on either end. Ben does not sit next to Mei. Siti sits somewhere to the left of Ali. Ben is not on the left end. In what order do they sit, from left to right?

Answer key (tutor only)

Lesson 1: Million Puzzles

S1. Four million, thirty thousand, two hundred and five.

S2. 600 000

S3. 5 483 000

S4. Greatest 6 543 210; smallest 1 023 456.

S5. 6 (each pack has 7 pens, 6 packs are 42 pens, 48 − 42 = 6). As one number sentence: 48 − 6 × (3 + 4) = 6.

S6. About 350 000 (7000 × 50). The exact answer is 331 776.

T1. 7 numbers: 1 000 001, 1 000 010, 1 000 100, 1 001 000, 1 010 000, 1 100 000 and 2 000 000.

T2. (2 + 3) × 4 + 5 = 25 and 2 + 3 × (4 + 5) = 29. (29 is the biggest you can make with one pair.)

T3. Range: 2 499 500 to 2 500 499. Six numbers: 2 500 002, 2 500 011, 2 500 020, 2 500 101, 2 500 110, 2 500 200. None below 2 500 000: 2 499 5__ already has digits adding to at least 29.

T4. 6 987 521 (12 479 below 7 000 000). The best number above is 7 125 689, which is 125 689 away.

T5. Greatest 640 499, smallest 639 500.

T6. 6 419 754 (7 654 321 − 1 234 567).

T7. 120 000, because 4000 × 30 = 120 000. (Exactly 119 132.)

T8. 29 = 5 × 7 − 2 × 3; 31 = 2 × 5 + 3 × 7; 100 = 2 × 5 × (3 + 7). Other ways are possible.

T9. (4 + 4 + 4) ÷ 4 = 3; (4 × 4 + 4) ÷ 4 = 5; 4 + 4 − 4 ÷ 4 = 7. Other ways are possible.

G1. Open-ended. Check the riddle by listing every number that fits the rounding clue first. Sample riddle with exactly one answer (checked on a computer over all 7-digit numbers): all digits different; rounds to 5 400 000 to the nearest hundred thousand; the ten thousands digit is 1 less than the hundred thousands digit; the thousands digit is the sum of those two digits; the tens digit is 0 and the hundreds digit is even; the digits add up to 30. Answer: 5 437 209.

G2. 64 × 15 625. (1 000 000 = 10 × 10 × 10 × 10 × 10 × 10; put all the 2s together to make 64 and all the 5s together to make 15 625.)

G3. 6 or 7. The smallest answer is 1000 × 100 = 100 000 (6 digits); the greatest is 9999 × 999 = 9 989 001 (7 digits).

G4. 1000. The first and last digits are 3; the 2nd, 3rd and 4th digits can be anything (10 × 10 × 10), and they fix the rest.

G5. (1 × 2 + 3) × 4 × 5 = 100 (or 1 × (2 + 3) × 4 × 5 = 100).

G6. About 10 days. 70 × 60 × 24 = 100 800 beats a day, and 1 000 000 ÷ 100 800 is just under 10.

G7. 231: (1 + 2) × (3 + 4) × (5 + 6) = 3 × 7 × 11. With no brackets it is only 33.

G8. 6 953 487. Seven different digits adding to 42 must be 3, 4, 5, 6, 7, 8, 9 (the biggest seven digits add to exactly 42), so there is no 0. The number is between 6 950 000 and 7 049 999, so it starts 6 9 with 5, 7 or 8 next; the smallest is 6 95 then 3, 4, 8, 7 (it must end in an odd digit).

G9. 2893. 9 × 1 + 90 × 2 + 900 × 3 + 4 = 9 + 180 + 2700 + 4. Go further: 5 888 896 (9 × 1 + 90 × 2 + 900 × 3 + 9000 × 4 + 90 000 × 5 + 900 000 × 6 + 7 = 9 + 180 + 2700 + 36 000 + 450 000 + 5 400 000 + 7).

Lesson 2: Fraction Feats

S1. 3/4 of a bar (cut every bar into 4 pieces: 12 pieces, 3 each; 3 ÷ 4 = 3/4).

S2. 3/4 eaten, 1/4 left.

S3. 4 + 3 = 7 slices eaten; 5 slices are left, which is 5/12 of the pizza.

S4. 9 (each group has 3 dots, and 3 groups make 9).

S5. 1/4 of the whole cake.

S6. Mei. Cut both bars into 12 equal parts: 3/4 = 9 parts and 2/3 = 8 parts. (Or: Mei is 1/4 short of a whole pizza and Ali is 1/3 short, and 1/4 is less than 1/3.)

T1. 1/3 + 1/24 and 1/4 + 1/8. Test 1/3, 1/4, 1/5, … in order: 1/5 leaves 7/40, not a unit fraction; after that the first fraction would be less than half of 3/8, so stop.

T2. 1/2. 2/3 was left after Ben; Ali ate 1/4 of it, so 3/4 of 2/3 = 1/2 is left.

T3. 1/20 and 1/90 (the bottom numbers multiply: 4 × 5 and 9 × 10).

T4. 4/6 = 2/3 of a litre.

T5. 6/7. 5/6 is 1/6 short of 1 and 6/7 is only 1/7 short of 1, and 1/7 is smaller than 1/6.

T6. 1/4 m (3/4 ÷ 3: 3 quarters shared by 3 pieces is 1 quarter each).

T7. 9/20 (2/5 = 8/20 and 1/2 = 10/20).

T8. 5/6 of 30. 7/8 of 24 = 21 and 5/6 of 30 = 25.

T9. 1/4 + 2/3 = 11/12 (only 1/12 away from 1). It is the closest of all 24 ways; 1/3 + 2/4 = 5/6 is 1/6 away.

G1. Only one way: 1/2 + 1/3 + 1/6. (The biggest part must be 1/2, and then 1/2 = 1/3 + 1/6 is the only split into two different unit fractions.)

G2. 1/100. Each top number cancels the bottom number before it, leaving 1 on top and 100 at the bottom.

G3. 120. What is left is 1/2 × 2/3 × 3/4 = 1/4 of the cookies, and 1/4 is 30.

G4. Open-ended. Sample: a shop had 60 oranges, sold 1/3 of them, then 1/4 of the oranges that were left. How many are left? (60 - 20 = 40; 1/4 of 40 = 10; 30 left.) Check that the model uses equal small units and that the second fraction is taken of the remainder.

G5. 8/35, from 3/7 − 1/5 (15/35 − 7/35).

G6. Exactly three ways: 1/2 + 1/5 + 1/20, 1/2 + 1/6 + 1/12 and 1/3 + 1/4 + 1/6. (The biggest part must be 1/2 or 1/3: three unit fractions of 1/5 or less add to less than 3/4. Then test in order, as in the lesson.)

G7. 5/14. Each fraction is a difference: (1/2 − 1/3) + (1/3 − 1/4) + (1/4 − 1/5) + (1/5 − 1/6) + (1/6 − 1/7). The middle parts cancel, leaving 1/2 − 1/7 = 5/14.

G8. Each child gets 5/12. Two ways: 1/3 + 1/12 (cut 4 pizzas into thirds and 1 pizza into twelfths), or 1/4 + 1/6 (cut 3 pizzas into quarters and 2 pizzas into sixths).

G9. (a) 54 litres: each hour 2/3 is left, 54 → 36 → 24 → 16. (b) 2 hours: 24 litres is less than half of 54 (27), but after 1 hour 36 litres is more than half. (c) 54 − 24 = 30 litres.

Lesson 3: Before and After

S1. Ali 14, Mei 10. The total is 24 before and 24 after.

S2. Ben 15 and his father 43. The father is 28 years older now, and 28 years older in 5 years (the difference stays the same).

S3. $30 (book $15, remainder $45, T-shirt $15)

S4. 4 (the total is 60, so 30 each)

S5. 10 (15 boys and 25 girls)

S6. 12 out of 36 = 1/3 blue; after the red beads are added, 12 out of 48 = 1/4 blue. The blue beads did not change.

T1. $120 (20 units: Monday 5 units, Tuesday 6 units; 11 units = $66, so 1 unit = $6)

T2. 70 at first, 100 now (the blue beads did not change: 4 units at first, and 2 parts of 5 after, so 1 part = 2 units and the whole box is 10 units; red 3 units at first and 6 units now, so 3 units = 30 and 1 unit = 10)

T3. 1 year ago (the difference, 32, is 4 units; 1 unit = 8: son 8, Mrs Tan 40)

T4. $80 (Ben 2 units, Siti 1 unit + $20; 3 units = $120, 1 unit = $40)

T5. 50 (the difference, 40, is 2 units; Mei 3 units = 60 after, so 50 at first)

T6. 72 litres (1/4 of the tank = 18 litres)

T7. 45 (Siti 3 units; after giving 1 unit she has 2 units, and Mei has 15 + 1 unit = 2 units, so 1 unit = 15)

T8. $15 (the difference, $30, is 1 unit; Ben $30, Ravi $60)

T9. 36 (the red pens did not change: red 1 unit, blue 2 units at first, 1 unit after; 1 unit = 12)

T10. 20 (the total, 80, stays the same and is 8 units, so 1 unit is 10; Ali ends with 30 and Ben with 50).

G1. Open-ended. Sample: Ali and Ben have 108 stickers altogether. Ali gives Ben 24 stickers. Then Ali has 1/2 as many stickers as Ben. How many did Ali have at first? (After: 3 units = 108, Ali 36, Ben 72; Ali had 36 + 24 = 60, Ben 48.) Check that the total is the same in both models and that every unit is the same size.

G2. Eight ways (Ali, Mei): 18 and 6 (Ali gives none), 16 and 8, 8 and 16, 6 and 18, 4 and 20, 3 and 21, 2 and 22, 1 and 23. The total stays 24, so the smaller number must go into 24 exactly and be less than 12: 1, 2, 3, 4, 6 or 8. (12 and 12 is equal, so it does not count.)

G3. 10 (in 5 years the total is 60 = 4 units, the son is 15)

G4. 160 (Ali 3 units, Ben 5 units; same means 4 units each, so Ben gave 1 unit = 20; 8 units = 160)

G5. 60 (the total did not change; 1/3 − 1/6 = 1/6 of the fruits is 10)

G6. $150 (15 units: dress 6 units, remainder 9 units, shoes 3 units; 3 units = $30, 1 unit = $10)

G7. $60 (Siti spent 1 unit, Mei spent 3 units; the $120 difference is 2 units, so 1 unit = $60; both have $20 left)

G8. 40 (work backwards: (4 + 4) × 2 = 16, (16 + 4) × 2 = 40)

G9. 40 (Ali 3 units, keeps 2 units; Ben keeps 2 units, so Ben had 4 units; 7 units = 140, 1 unit = 20)

Lesson 4: Triangle Areas

S1. 12 cm² (the rectangle is 24 cm² and each triangle is half of it).

S2. 32 cm². The two side pieces cover the middle triangle exactly, so the triangle is half of the square (64 ÷ 2).

S3. 6 cm² (the rectangle is 4 × 3 = 12 cm² and one triangle is half of it).

S4. 10 cm², and it does not change when the point moves along the top side (the triangle is always half of the rectangle).

S5. 30 cm² (each triangle is 15 cm²).

S6. 12 cm² for everybody (half of 24), wherever the top corner is.

T1. 9 cm

T2. 96 cm² (1/2 × 12 × 16)

T3. 8 cm (the square is 36 cm², and 1/2 × 9 × 8 = 36).

T4. 42 cm² (84 – 42)

T5. 10 cm (area 20 cm²)

T6. 48 cm² and 48 cm²

T7. 48 cm² (36 + 12)

T8. 24 cm (Ali's area is 30 cm², Mei's is 60 cm², and 1/2 × 5 × 24 = 60).

G1. Open-ended. Sample: ABCD is a rectangle 10 cm by 6 cm, E is the middle of the top side AB and F is the middle of the right side BC. The shaded triangle is DEF (D is the bottom-left corner). Taking away: 60 – 15 (ADE) – 7.5 (EBF) – 15 (FCD) = 22.5 cm². Check by a second method, for example by drawing the puzzle on 1-cm grid paper and counting squares, that the answers agree.

G2. 36 cm² (96 – 24 – 12 – 24)

G3. The first triangle has area 30 cm². (a) 90 cm² (3 times as big). (b) 45 cm² (3 times as big, then half). (c) The height must be 1/4 as long, 1.25 cm. (d) The area is multiplied by the same numbers as the base and the height, one after the other: × 3 and × 1/2 gives × 1 1/2.

G4. 16 cm² (add the shape ABCF to both: square 64 cm² and triangle ABE 48 cm², so the difference is 64 – 48)

G5. 6 pairs: 1 and 12, 2 and 6, 3 and 4, 4 and 3, 6 and 2, 12 and 1 (base × height must be 12)

G6. (a) 84 cm² (1/2 × 14 × 12). (b) 11.2 cm (1/2 × 15 × height = 84, so height = 168 ÷ 15). (c) The area is fixed, so a longer base needs a shorter height.

G7. 12 cm (area 1/2 × 15 × 20 = 150 cm², and 1/2 × 25 × 12 = 150)

G8. 50 cm² both times: it is always half of the big square (box it in a rectangle and take away three right-angled triangles: 160 – 50 – 12 – 48 = 50, and 140 – 50 – 12 – 28 = 50)

G9. 18 cm² (the height is 9 cm because 1/2 × 12 × 9 = 54; ABD has base 4 cm and the same height)

Lesson 5: Volume Vault

S1. 6 cubes in the bottom layer, 2 layers, 12 cubes altogether (12 cm³).

S2. 3 rectangles: 1 by 12, 2 by 6 and 3 by 4.

S3. 300 cm³; 1000 cm³, which is 1 ℓ.

S4. 60 cubes (a layer of 20, three layers).

S5. 1 cm (12 cubes fill exactly one layer of 4 × 3).

S6. 2000 cm³; 2500 ml.

T1. 24 (4 along, 3 across, 2 layers; not 270 ÷ 8, because the cubes cannot be cut)

T2. 45 ℓ (3/5 – 1/3 = 4/15 of the tank is 12 ℓ, so 1/15 is 3 ℓ)

T3. 7 cm (base area 245 ÷ 5 = 49 cm², and 7 × 7 = 49)

T4. 6 (the inside block is 1 × 2 × 3)

T5. 12 buckets (3/4 of 120 ℓ = 90 ℓ; 90 ÷ 8 = 11.25, so 11 buckets are not enough)

T6. 30 cm (36 000 ÷ 1200)

T7. 5 cm (8000 ÷ 1600)

T8. 11.5 cm (3 ℓ = 3000 cm³ raises the water 3000 ÷ 1000 = 3 cm to 11 cm; the stone raises it 500 ÷ 1000 = 0.5 cm more).

T9. 3000 cm³ (40 × 25 × 3)

T10. 8 cm (14 400 cm³ of water spreads over both bases, 600 + 1200 = 1800 cm², so 14 400 ÷ 1800 = 8)

G1. 216 (each face has 96 ÷ 6 = 16 = 4 × 4 one-face cubes, so the cube is 6 cubes along each edge: 6 × 6 × 6)

G2. 7 (1 × 1 × 64, 1 × 2 × 32, 1 × 4 × 16, 1 × 8 × 8, 2 × 2 × 16, 2 × 4 × 8, 4 × 4 × 4)

G3. 6600 cm³ = 6.6 ℓ (40 × 30 × 6 − 10 × 10 × 6 = 7200 − 600; the cube is taller than the water, so only 6 cm of it is under water)

G4. 22.5 cm (each brick raises it 3 ÷ 4 = 0.75 cm; 10 bricks raise it 7.5 cm)

G5. Open-ended. Sample: a tank has a base 25 cm by 20 cm and water 10 cm deep. A stone of 1000 cm³ sinks. How much does the water rise? (1000 ÷ 500 = 2 cm, so the water is 12 cm deep.) If the base were twice as wide (1000 cm²), the rise would be half as much, 1 cm.

G6. 72 (the smallest cube edge is 12 cm, the smallest common multiple of 2, 3 and 4: 6 × 4 × 3 = 72 bricks)

G7. 60 cm³ (edges 3 cm, 4 cm and 5 cm: 3 × 4 = 12, 3 × 5 = 15, 4 × 5 = 20)

G8. 24 with 2 faces (4 × 1 + 4 × 2 + 4 × 3 along the edges); 22 with 1 face (60 − 8 corners − 24 − 6 inside)

G9. 12 cm (the water now covers 600 − 100 = 500 cm² of floor: 6000 ÷ 500 = 12 cm; the block is 15 cm tall, so it sticks out. Not 12.5 cm!)

Lesson 6: Rate Racers

S1. $4; $6; 50¢ each.

S2. 15 minutes

S3. Pack A. Two Pack A cost $4 for 6 pens, but Pack B costs $5 for 6 pens. (Pack A is about 67¢ a pen and Pack B is about 83¢ a pen.)

S4. 15 minutes

S5. 5 litres after 1 minute, 10 after 2 minutes, 25 after 5 minutes. 20 litres takes 4 minutes (5 + 5 + 5 + 5).

S6. $9.75

T1. 5 ℓ a minute (the tank holds 180 ℓ; 180 ÷ 9 = 20 ℓ a minute together; 20 − 15 = 5)

T2. $9.60 (one box of 10 and three boxes of 6 make 28 eggs; three boxes of 10 cost $9.90)

T3. $9.50 (4 h 35 min: first hour, then 3 h 35 min is 8 half hours or parts)

T4. 100 minutes (Plan B costs 15¢ more each minute; $15 ÷ 15¢ = 100; both cost $25)

T5. 900 (1 machine makes 30 masks a minute: 30 × 5 × 6)

T6. Shop Y by $4. At X, 20 pens cost $16.00. At Y, each deal gives 5 pens for $3.00, so 20 pens are 4 deals, $12.00.

T7. 30 minutes (300 ÷ 10)

T8. 216 ℓ (in 1 day there are 86 400 seconds, 43 200 drips of 5 ml is 216 000 ml).

T9. $13.64: 1 deal from C (12 cans, $7.80) + 2 packs from B (8 cans, $5.84). The only other ways are 2 packs from A + 2 packs from B = $14.24 and 5 packs from B = $14.60.

T10. 540 pages (together they print 50 pages a minute, so the job takes 18 minutes; P prints 30 × 18 = 540 and Q prints 20 × 18 = 360)

T11. 20 minutes (half is 120 ℓ, and 120 ÷ 6 = 20).

T12. 3 hours ($3.60 − $1.20 = $2.40 = 4 half hours after the first hour, so up to 1 h + 2 h = 3 hours).

G1. 2 minutes (pretend the job is 6 pages: P prints 1 page a minute, Q 2 pages, together 3 a minute)

G2. 30 pens (6 pens cost $4 and sell for $6: $2 profit for every 6 pens)

G3. 10.15 a.m. (400 cups by 10.00; 1200 ÷ 80 = 15 minutes more)

G4. $13.64 (1 deal from C and 2 packs from B make 20 cans). Check the bigger totals: 22 cans cost at least $14.92 (C + A + B) and 24 cans cost at least $15.60 (2 deals from C), so they cannot be cheaper.

G5. 15 minutes (135 000 cm³ = 135 ℓ; 135 ÷ 9)

G6. Open-ended. Sample: 6 for $3.00 (50¢ each), 10 for $5.50 (55¢ each) and 12 for $7.20 (60¢ each): the 6-pack is the best buy. If the 12-pack cost $5.40 or less (45¢ each or less) it would be the best buy.

G7. 3 cats (each cat eats 1 fish every 3 minutes, so the 3 cats eat 1 fish a minute: 100 fish in 100 minutes)

G8. After 6 hours, both 8 cm tall (the gap of 6 cm closes by 1 cm each hour)

G9. 30 minutes (pretend the bath holds 60 ℓ: 5 ℓ a minute in, 3 ℓ a minute out, 2 ℓ a minute stay: 60 ÷ 2)

Lesson 7: Percentage Puzzles

S1. 35% shaded, 65% not shaded.

S2. 50% is $40, 25% is $20 and 10% is $8.

S3. $5 is taken off; you pay $45.

S4. 60% (24 out of 40 = 60 out of 100)

S5. $45

S6. GST is 9 × 20¢ = $1.80; you pay $21.80.

T1. (a) 25% ($1 out of $4) (b) 20% ($1 out of $5) (c) Each change is compared with its own starting amount: $1 out of $4 is a bigger part than $1 out of $5. (This is a puzzle that goes further than the lesson.)

T2. (a) Shop X $45 (she pays for 3 boxes); Shop Y $42 (30% of $60 = $18 off). Shop Y is the better buy, by $3. (b) 25% off (1 box free out of 4, $15 off $60)

T3. She found 10% ($8) and 5% (half of 10%, $4) and added them. 15% of $46 = $4.60 + $2.30 = $6.90.

T4. Saved $30 (10% = $25, 2% = $5); sale price $220.

T5. $12.80 (GST $7.20, so the watch costs $87.20)

T6. $60 (shoes $80, $120 left, bag $60, $60 left)

T7. They are the same: both are 21. In general, a% of b is the same as b% of a (for example 4% of 25 = 25% of 4 = 1).

T8. Test A: 72% (18 out of 25) is more than 67.5% (27 out of 40).

G1. Open-ended. Sample: a $60 jacket goes up 50% and then down 50%. Ravi says the price is back to $60. Really: $60 + $30 = $90, then 50% of $90 is $45 off, so $45. It is 25% LESS than the start, because the second 50% is of the bigger amount.

G2. No. $40 is 80% of the old price, so 10% is $5 and 100% is $50. (Ben added 20% of $40, but the 20% was of the old price.)

G3. 200 ℓ (the 60 ℓ is 30% of the tank, so 10% is 20 ℓ)

G4. 50. Work backwards: 48 is 80% of Test 2, so Test 2 = 60; 60 is 120% of Test 1, so Test 1 = 50.

G5. 25.5% (boys 360, 72 take the bus; girls 440, 132 take the bus; 204 out of 800)

G6. 10. The 5 blue marbles stay. If they are 50% of the box, there are 10 marbles left, so 20 − 10 = 10 red marbles are taken out.

G7. No. $109 is 109% of the price before GST, so 1% is $1 and the GST is $9. (9% of $109 is wrong because GST is 9% of the price BEFORE GST.)

G8. Shop P, by $1.20. P: $80 − $20 = $60. Q: $80 − $8 = $72, then 15% of $72 = $10.80 off, $61.20.

G9. $120. Work backwards: $54 is 75% of the remainder, so the remainder is $72. $72 is 60% of her money, so 10% is $12 and 100% is $120.

Lesson 8: Angle Chasers

S1. The three corners fit together along a straight line, so the three angles add up to 180°.

S2. 53° (180° − 127°).

S3. 45° for one slice (360° ÷ 8) and 90° for two slices.

S4. About 65° (180° − 48° − 67°). The three angles add up to 180°.

S5. 45°, then 22.5° (half of 45°).

S6. 360° (a full turn). For the rectangle, 4 × 90° = 360° too.

T1. A = 135° and B = 45°. Adjacent angles of a parallelogram add up to 180°, so A and B make 4 equal parts of 45°: B = 45°, A = 135°.

T2. 35° and 35°. 110° cannot be a base angle (two of them would be 220°), so it is the top angle and 180° − 110° = 70° is shared by two equal base angles.

T3. AEC = 118° and EAB = 62°. ADC = 56° (opposite angles); DEA = (180° − 56°) ÷ 2 = 62° (DA = DE); AEC = 180° − 62° = 118° (straight line DC); DAB = 180° − 56° = 124° (neighbours in a parallelogram), so EAB = 124° − 62° = 62°.

T4. ADE = 150° (90° + 60°). Triangle ADE is isosceles (AD = DE), so DAE = (180° − 150°) ÷ 2 = 15°.

T5. ADC = 96° and BCD = 83° (angles between the parallel sides add up to 180°). Check: 84° + 97° + 96° + 83° = 360°.

T6. 40° (AB = BC, so triangle ABC is isosceles: (180° − 100°) ÷ 2)

T7. 150° at 5 o'clock (5 × 30°). At 8 o'clock the big angle is 240°, so the smaller angle is 120°.

T8. 60° (a full turn of 360° takes 60 minutes, so 6° a minute) and 270°.

T9. A = 110°, B = 70°. Adjacent angles add up to 180°; 180° − 40° = 140°, and 140° ÷ 2 = 70°, so A is 70° + 40° = 110°.

T10. 142°, 38°, 142°. There are two different sizes: 38° and 142°.

G1. 30°, 30°, 120°. The two right angles make a straight line, and the two 60° angles together make 120°.

G2. Open-ended. Sample: ABCD is a parallelogram with angle ABC = 56°, and E is on DC with DA = DE. Find angle EAB. (ADC = 56°, DEA = (180° − 56°) ÷ 2 = 62°, DAB = 180° − 56° = 124°, EAB = 124° − 62° = 62°.) Check that every step has a reason and that the given angles do not contradict each other.

G3. Angle A = 60°, so each part is 30°. ADB = 180° − 70° − 30° = 80°, and ADC = 180° − 80° = 100° (check: 30° + 50° + 100° = 180°).

G4. 20°, 20°, 140° and 20°, 80°, 80°.

G5. Triangle ACD: 20°, 40°, 120°. Triangle BCD: angle BCD = 60°, B = 60° and D = 60°, so it is equilateral.

G6. Angle BAD = 108° (adjacent angles add up to 180°). Triangle ABE is isosceles with top angle 108°, so ABE = 36°. Then EBC = 72° − 36° = 36°.

G7. 30°. Angles B and C are (180° − 40°) ÷ 2 = 70°. In triangle BCD, BD = BC, so angle BDC = angle C = 70°, which makes angle CBD = 40°. Then ABD = 70° − 40° = 30°.

G8. 150°. AE = AB = AD, so triangle ADE is isosceles with angle DAE = 90° − 60° = 30°, which makes angle AED = 75°. In the same way angle BEC = 75°. Angle AEB = 60°. Round E: 360° − 75° − 75° − 60° = 150°.

G9. Two sets: 50°, 50°, 80° and 40°, 70°, 70°. (If the top angle is 30° bigger than a base angle: 3 equal parts and 30° make 180°, so base 50°. If a base angle is 30° bigger than the top: top 40°, base 70°.)

Lesson 9: Pattern Generalisers

S1. 41 (5 + 9 × 4: there are 9 jumps of 4 from the 1st number to the 10th).

S2. 25 (3 × 8 + 1).

S3. Yes. 50 − 2 = 48 and 48 ÷ 4 = 12 jumps, so 50 is the 13th number.

S4. 7 tricycles (suppose all 15 are bicycles: 30 wheels; 7 wheels short; each tricycle adds 1 more wheel). 8 bicycles. Check: 16 + 21 = 37.

S5. 47

S6. 4 files and 6 pens (all pens would be $20; each file instead of a pen adds $1). Check: $12 + $12 = $24.

T1. Red (50 = 8 × 6 + 2, so it is the 2nd bead of a group). 49 yellow beads: 16 full groups of 6 give 96 beads and 48 yellow, and beads 97 to 100 are red, red, blue, yellow.

T2. 96 (each row of 5 seats 12, and 8 × 12 = 96). One long row seats only 82. Splitting the tables gives more end seats.

T3. 12 goats and 23 chickens (all chickens would have 70 legs; 24 legs short; each goat adds 2 legs). Check: 48 + 46 = 94.

T4. 4, 9, 16: square numbers. The first 20 odd numbers add up to 20 × 20 = 400. (Each new odd number adds an L-shaped layer to a square of dots.)

T5. $31 ($3 + 14 × $2).

T6. 203 (the 41st number: 3 + 40 × 5).

T7. 26 and 27. 25 × 26 = 650 is too small and 27 × 28 = 756 is too big. Check: 26 × 27 = 702.

T8. 7 (all correct would be 30; each wrong answer costs 3 + 1 = 4 marks; 12 ÷ 4 = 3 wrong). Check: 21 − 3 = 18.

G1. 55 (1 + 2 + 3 + ... + 10). Two copies of the staircase make a 10 by 11 rectangle: 110 ÷ 2 = 55.

G2. Open-ended. Sample: matchstick triangles in a row use 3, 5, 7, ... sticks, so sticks = 2 × figure number + 1. Figure 50 uses 101 sticks. No figure uses exactly 100 sticks, because 2 × figure number + 1 is always odd (figure 49 uses 99 and figure 50 uses 101). Check that the rule works for figures 1, 2 and 3.

G3. 30 (16 small, 9 of size 2 by 2, 4 of size 3 by 3 and 1 of size 4 by 4).

G4. (a) 8 × 8 + 12 × 6 = 136 legs. (b) The smallest total is 120 (no spiders) and each spider instead of a beetle adds 2 legs, so the totals are every EVEN number from 120 to 160: 21 different totals. (c) 135 is odd and every total is even. 148 = 120 + 28, so 14 spiders (and 6 beetles).

G5. 342 (9 one-digit pages, 90 two-digit pages = 180 digits, 51 three-digit pages = 153 digits).

G6. 220. There are 11 rows of 10 sticks going across and 11 columns of 10 sticks going down: 110 + 110.

G7. 61. The counts go 1, 5, 13, 25, 41, 61 (add 4, 8, 12, 16, 20). Or: each diamond is two square numbers of dots, 6 × 6 + 5 × 5 = 36 + 25 = 61.

G8. 18. She answered 22 questions. All 22 correct would be 88 marks; each wrong answer costs 4 + 1 = 5 marks; 88 − 68 = 20, 20 ÷ 5 = 4 wrong. Check: 72 − 4 = 68.

G9. 5. The numbers 1 to 9 use 9 digits. That leaves 91 digits: 45 two-digit numbers (10 to 54) use 90 of them, so the 100th digit is the first digit of 55.

Lesson 10: Challenge Day

S1. 14 (work backwards: 50 + 6 = 56, 56 ÷ 4 = 14).

S2. $18 ($9 is the other half).

S3. 24 and 16 (take away the 8: 32, share: 16 and 16, then add 8 back to one: 24 and 16).

S4. Siti. Ravi has the rabbit (no bird, no hamster). Siti is not the hamster, so she has the bird, and Ben has the hamster.

S5. 5 (4 small and 1 big).

S6. 4 pads and 6 pens (all pens would be $20; each pad instead of a pen adds $1).

T1. $60. Work backwards: $24 is 2/3 of what was left, so that was $36; $36 is 3/5 of her money, so 1/5 is $12 and she started with $60. Check: 60 − 24 = 36, 36 − 12 = 24.

T2. 15 numbers: 104, 113, 122, 131, 140, 203, 212, 221, 230, 302, 311, 320, 401, 410, 500.

T3. 72 and 73 (145 is 1 more than twice 72).

T4. Day 8. After 7 days and nights it is at 7 m. On day 8 it climbs 3 m to reach 10 m before it can slip.

T5. 7 fifty-cent coins and 7 twenty-cent coins (all 20¢ would be $2.80; $2.10 short; each swap adds 30¢; 210 ÷ 30 = 7).

T6. 61 (it is 1 more than a multiple of 2, 3, 4, 5 and 6; the smallest such multiple is 60).

T7. 12. Work backwards: 22 − 3 = 19, 19 × 2 = 38, 38 + 10 = 48, 48 ÷ 4 = 12.

T8. 30. The difference between a half and a third is 1/6, and 1/6 of the number is 5. Check: 15 and 10.

G1. 27. Work backwards: 12 − 3 = 9; 9 is half of what was left, so 18 were left after stop 1; 18 is 2/3 of the passengers, so 1/3 is 9 and there were 27 at first.

G2. 15 (5 of length 1, 4 of length 2, 3 of length 3, 2 of length 4 and 1 of length 5: 5 + 4 + 3 + 2 + 1).

G3. 60 rectangles. Count by size, width by height (the grid is 4 wide and 3 tall): 1 by 1: 12, 2 by 1: 9, 3 by 1: 6, 4 by 1: 3 (30 with height 1); 1 by 2: 8, 2 by 2: 6, 3 by 2: 4, 4 by 2: 2 (20 with height 2); 1 by 3: 4, 2 by 3: 3, 3 by 3: 2, 4 by 3: 1 (10 with height 3). 30 + 20 + 10 = 60. Check: a rectangle is chosen by 2 of the 5 vertical lines (10 ways) and 2 of the 4 horizontal lines (6 ways): 10 × 6 = 60.

G4. 15 (in the ones place: 4, 14, 24, 34 and 44 is 5 times; in the tens place: 40 to 49 is 10 times; 5 + 10 = 15).

G5. Open-ended. Check by trying to find a second solution: if one exists, add a clue. A good test is to remove each clue in turn and count the solutions. Sample: the hawker-centre puzzle in Question 2 has exactly one solution (checked by trying all 24 arrangements); without clue 4 (Mei did not order roti prata) it has more than one.

G6. 14. Work backwards: after day 3 there are 0; before day 3 there were 2 (half of 2 is 1, and 1 more is 2); before day 2: (2 + 1) × 2 = 6; before day 1: (6 + 1) × 2 = 14. Check: 14 → 6 → 2 → 0.

G7. 45 litres, which is 3/8 of the tank. The 20 litres are 1/2 − 1/3 = 1/6 of the tank, so the full tank holds 120 litres. Half full is 60 litres; letting out 1/4 of it (15 litres) leaves 45 litres, and 45 ÷ 120 = 3/8.

G8. 3 fifty-cent coins, 6 ten-cent coins and 16 twenty-cent coins. Suppose all 25 coins were 20¢: that is $5.00. Every 50¢ coin comes with two 10¢ coins; this group of 3 coins is worth 70¢, and replacing three 20¢ coins (60¢) with it adds 10¢. The extra is 30¢, so there are 3 groups. Check: 150¢ + 60¢ + 320¢ = 530¢ and 3 + 6 + 16 = 25.

G9. Siti, Mei, Ali, Ben. Mei is in chair 2 or 3. If Mei is in chair 3, the only chair not next to her is chair 1, but Ben is not on the left end, so that fails. So Mei is in chair 2, the only chair not next to her is chair 4, so Ben is in chair 4. Siti and Ali are in chairs 1 and 3 and Siti is on the left: Siti 1, Ali 3.